Experiments from Start to End

Local question answering on long and educational videos · Sayed Ilham Azhar Harun · project *local-surf* · October 2026 · one 16 GB GPU, small frozen open models

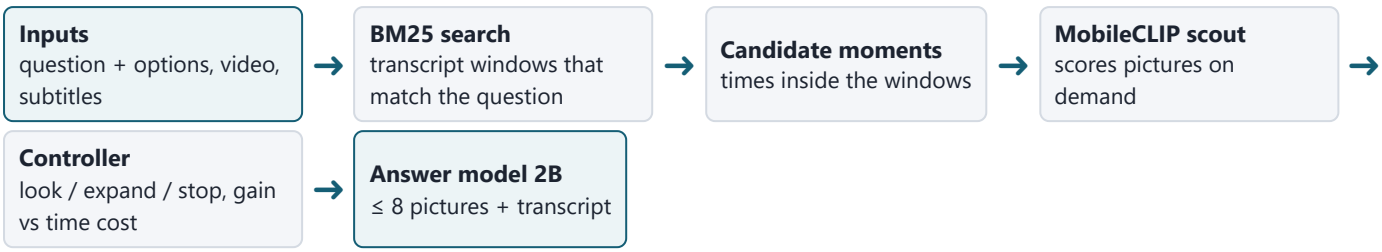
For every experiment: first the **structure** of the framework that was tested (a flow diagram, read left to right), then the **performance table**. Section 10 is the full grid of every framework version, with every answer model and with or without speech text.

How to read this document

Accuracy	share of questions answered correctly (the final score). Every percentage is accuracy unless the table says otherwise.
Evidence hit	share of questions where the chosen pictures include the moment human annotators marked as showing the answer (CG-Bench only). It measures *looking in the right place*, not answering.
Dev questions	used during the research to choose settings. Numbers on dev guided decisions.
Test questions	locked away until the end and used once. Test numbers are the trustworthy results.
Confirmed	the 95% confidence interval of a difference does not include zero (bootstrap over videos).
Answer model	the frozen vision-language model that reads the pictures and answers: Qwen3-VL-2B (small) or Qwen3-VL-4B (larger). Never trained by us.
Speech text	up to 600 transcript words given to the answer model next to the pictures. "No speech text": the transcript is still used to *find* pictures, but no words are shown to the answer model.
Head A	our trained speech-path model: decides *where to look* from the spoken words.
Head B	our trained picker: decides *which 4 candidate pictures* to show the answer model.

Dataset	What it is	Used for
LongVideoBench	public benchmark (NeurIPS 2024): 3,763 videos, multiple choice, subtitles; we used 440 validation questions	v1
MIT lecture questions (self-made)	197 multiple-choice questions generated by a larger model from 20 MIT OpenCourseWare lectures, machine-checked; answer-position bias found and corrected	v2
EduVidQA	public lecture benchmark (EMNLP 2025), open-ended answers	v3 check only
Video-MMMU	public lecture benchmark (2025): 300 university lecture videos, 10-option multiple choice. **Perception** = read or see what is on the slide; **Comprehension** = understand what was taught	v3 (lectures)
CG-Bench	public long-video benchmark (2024): videos of 10–80 min, human-marked answer seconds; we used 51 English-subtitled videos (train 37 / dev 8 / test 6)	v3 (long everyday videos), head training

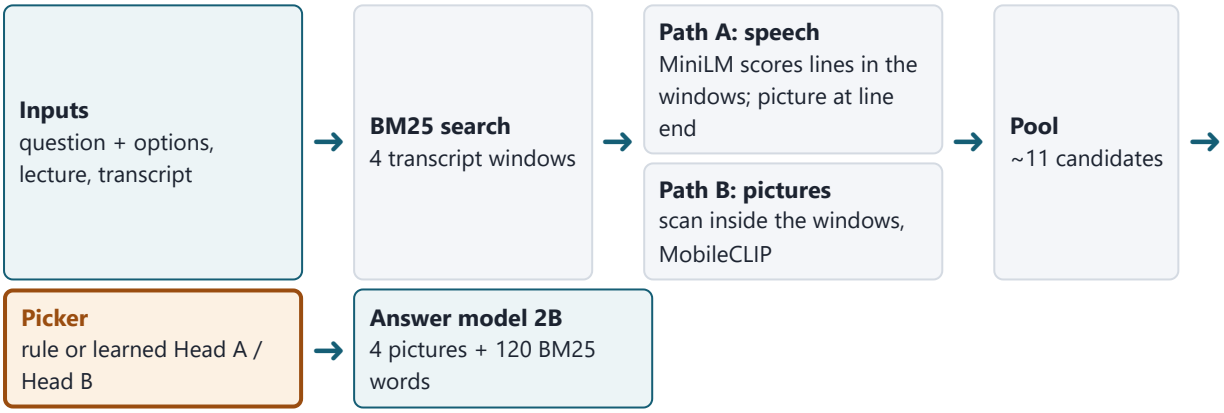
1 · Version 1: cost-aware controller (LongVideoBench)



Frame strategy (440 questions)	Multiple choice	Open-ended (local judge, not human-checked)	Time per question
Transcript only (no pictures)	32.5	44.1	1.14 s
Evenly spaced pictures + transcript	39.3	33.4	1.26 s
Transcript-retrieved moments	36.4	32.0	1.11 s
MobileCLIP top pictures	**41.6**	**32.3**	**2.25 s**
Cost-aware controller	37.5	29.8	1.47 s

Research idea (train the controller to look when important speech is damaged): not supported; damage changed the answer in only 13% (train) / 4% (dev) of cases. Chance level 21.4%.

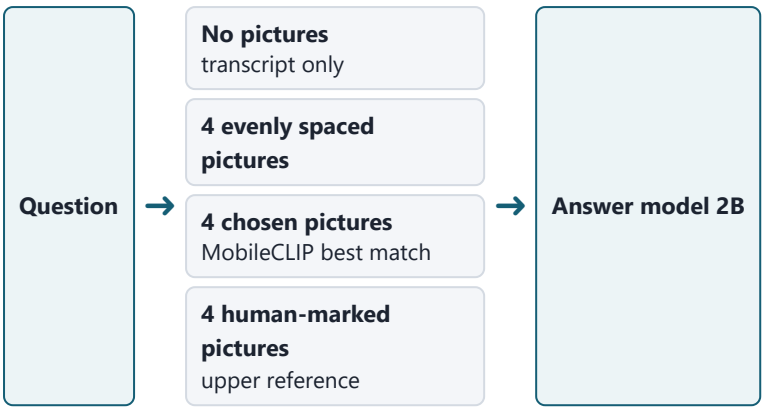
2 · Version 2: two candidate paths and first learned heads (MIT lectures)



Picker (197 questions, 20 lectures, each tested by models that never saw it)	Accuracy
Transcript only	15.2
MobileCLIP top-4	34.5
MobileCLIP + diversity	36.0
Learned Head B (with screen text)	34.7
Learned Head A frame credit	30.5 (confirmed worse)
Zero-shot transcript relevance (no training)	**38.6**

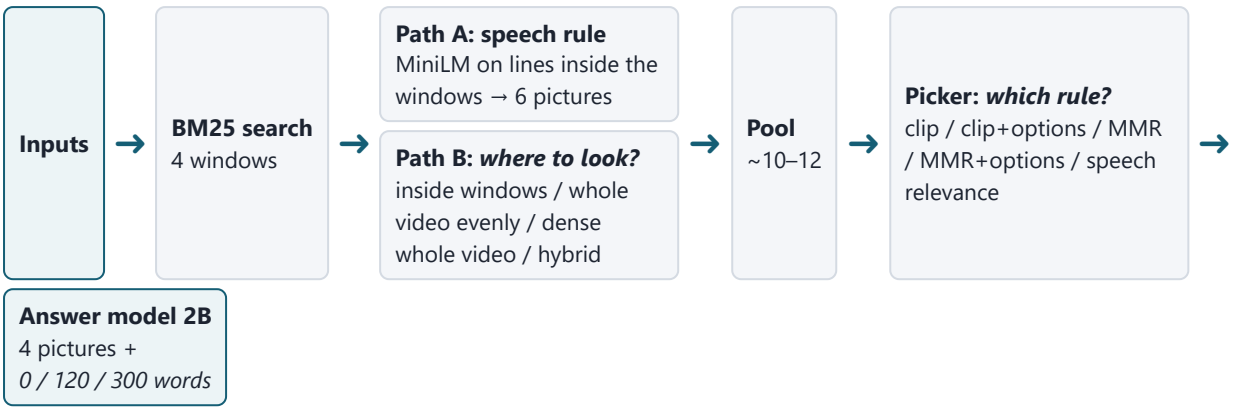
Follow-up	Result
Learned "4th picture" completion	−1.0 (refuted)
Spread the picture scan over all windows	34.5 → 39.1 (+4.6, not confirmed)
Same on LongVideoBench (transfer check)	43.0 → 42.5 (no gain)
Option-rotation scoring (fixes answer-position bias)	MobileCLIP top-4: 49.7 rotated vs 34.3 transcript only

3 · v3 benchmark check: can pictures change the answer?



Benchmark (dev)	No pictures	Evenly spaced	Chosen pictures	Human-marked pictures	Decision
EduVidQA (open-ended, FactQA score)	0.587	0.493	0.498	—	pictures do not help: not used for picture claims
Video-MMMU Perception	40.2	52.6	56.7	—	pictures help (+16.5, confirmed)
CG-Bench	15.9	17.1	40.2	63.4	pictures help (+24.4); large room

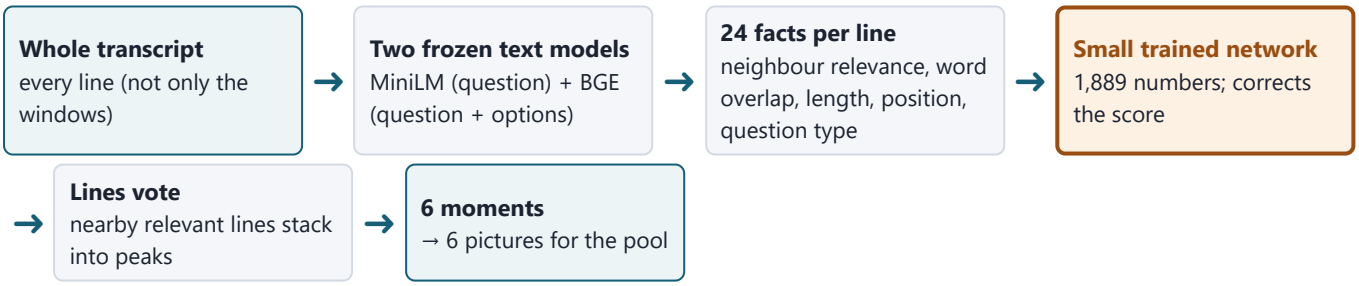
4 · v3 tuning of the rule-based framework (dev only)



Setting (2B, 4 pictures)	Long videos (CG-Bench, 82)	Lectures screen-reading (98)	Right picture among candidates (CG-Bench)
Starting framework v2: Path B inside windows, 120 BM25 words	28.0	56.1	33%
Path B scans the whole video evenly	34.1	57.1	50%
Path B dense whole-video scan	37.8	52.0	62%
Path B hybrid (half windows, half dense) + MMR on question+options	35.4	58.2	60%
... + no transcript words (frozen best framework)	**40.2**	**63.3**	**60%**
... with 2 / 8 pictures	37.8 / 39.0	52.0 / 60.2	

Best framework vs starting framework on dev: +12.2 on long videos (confirmed).

5 · Head A: trained speech-to-moment locator

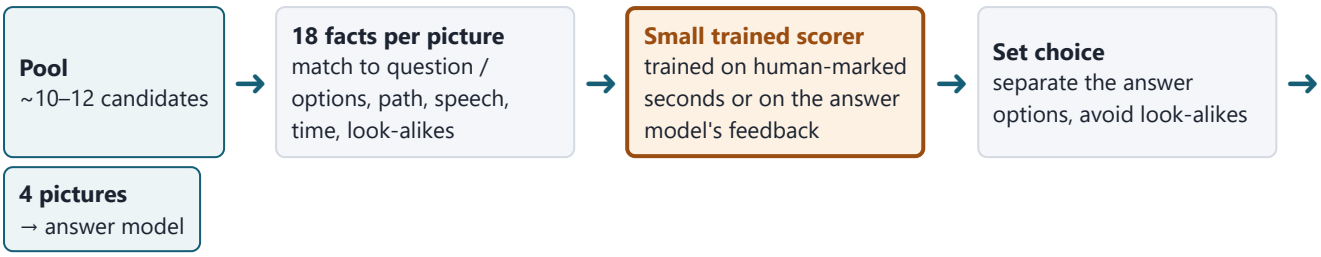


Trained on 877 CG-Bench questions (83 videos) whose answer seconds were marked by humans; measured on 164 new dev questions. Number = evidence hit: at least one of 6 moments inside the marked seconds.

Head A method	Evidence hit (6 moments)
Zero-shot MiniLM, lines inside windows (speech rule of the best framework)	25.0
Zero-shot MiniLM, whole transcript	28.0
Fine-tuned MiniLM	~8 (training destroyed its ranking)
Larger zero-shot rerankers (BGE up to 568M)	up to 31.1
Light trained network on frozen scores	32.7
... with learned speech-to-picture offsets (original idea)	31.7
... lines vote + BGE facts, 3 copies averaged (chosen Head A)	**36.6 (+8.5, confirmed)**

Inside the best framework (2B, no speech text, dev)	Right picture among candidates	Accuracy long videos	Accuracy lectures screen-reading
Speech rule (best framework)	59.8	40.2	63.3
Head A	64.6 (+4.9, confirmed)	39.0	62.2

6 · Head B: trained picker of the final 4 pictures



Trained on 373 CG-Bench questions; the feedback version used 4,203 answers given with one picture at a time. Numbers: accuracy difference to the MMR rule, pooled over 278 dev questions (long videos + both lecture tracks), 2B, no speech text.

Head B method	Difference to the MMR rule
Option-evidence set rule (no training, original idea)	−1.4
Trained on human-marked seconds	−1.4
... + option-evidence term (chosen Head B, "ev_set")	**−0.4**
Trained on answer-model feedback	−3.6
Confidence choice: answer with 4 picture sets, keep the most confident	+1.1 (not confirmed)

7 · Final locked test (declared in advance, run once; 2B, no speech text except v2)

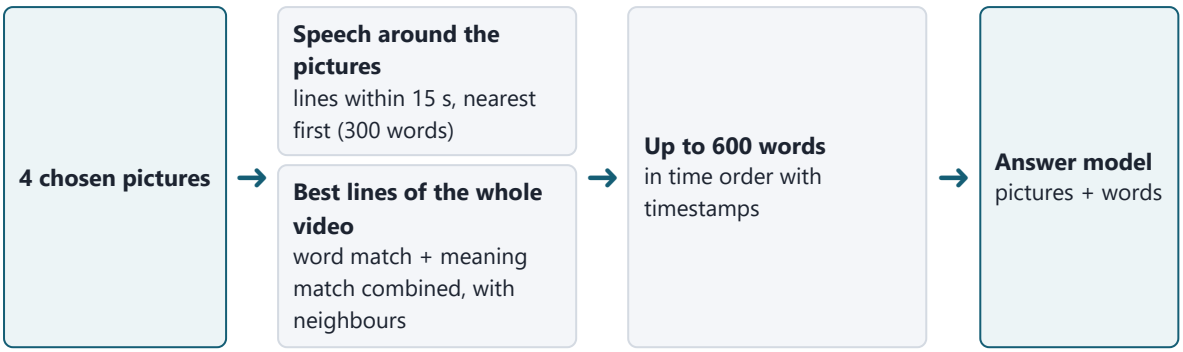


System	Long videos (55)	Lectures screen-reading (202)	Lectures understanding (202)	All 459
No pictures	20.0	25.2	19.8	22.2
4 evenly spaced pictures (standard rule)	32.7	40.6	33.2	36.4
Dense MobileCLIP top-4 (standard rule)	41.8	40.6	31.7	36.8
Starting framework (v2)	34.5	47.0	30.2	38.1
Best framework	**50.9**	**48.0**	**32.2**	**41.4**
Best + Head A	41.8	45.0	33.7	39.7
Best + Head B	47.3	48.0	33.2	41.4
Best + Head A + Head B	34.5	52.0	32.7	41.4
Best + confidence choice	47.3	50.0	33.2	42.3

Difference in points (95% interval)	Long videos	Screen-reading	Understanding	All 459
Best − starting framework	+16.4 (+7.0 to +25.9)	+1.0 (-5.0 to +6.9)	+2.0 (-3.0 to +6.9)	+3.3 (-0.7 to +7.0)
Head A − best	-9.1 (-15.7 to -3.4)	-3.0 (-7.4 to +1.5)	+1.5 (-2.5 to +5.4)	-1.7 (-4.4 to +1.1)
Head B − best	-3.6 (-11.5 to +4.1)	+0.0 (-5.0 to +4.5)	+1.0 (-3.5 to +5.4)	+0.0 (-3.0 to +3.1)
Head A+B − best	-16.4 (-19.6 to -13.0)	+4.0 (-0.5 to +8.4)	+0.5 (-4.0 to +5.0)	+0.0 (-3.1 to +3.3)
Confidence choice − best	-3.6 (-7.7 to +0.0)	+2.0 (-2.0 to +5.9)	+1.0 (-3.0 to +5.0)	+0.9 (-1.6 to +3.5)
Best − dense MobileCLIP	+9.1 (+1.9 to +15.6)	+7.4 (+2.5 to +12.9)	+0.5 (-4.0 to +5.4)	+4.6 (+1.3 to +7.8)
Best − evenly spaced	+18.2 (+8.8 to +27.3)	+7.4 (+1.5 to +13.9)	-1.0 (-6.4 to +4.5)	+5.0 (+1.1 to +8.9)

Green / red = confirmed gain / loss. The long-video test has only 6 videos, so its intervals are unreliable.

8 · Speech text for the answer model: keeping the important words



Text given to the answer model (long videos, dev)	Important speech kept	Open-ended correct	Multiple choice
Starting framework v2 (BM25, 120 words)	28%	13.4	28.0
Best pictures + BM25 120 words	28%	17.1	35.4
Best pictures, no words	0%	19.5	40.2
Best pictures + whole transcript (capped 1,500 words)	66%	19.2	40.2
Best pictures + meaning-based search, 300 words	62%	20.7	37.8
Best pictures + speech around pictures + search, 600 words (chosen)	**68%**	**22.0**	**37.8**

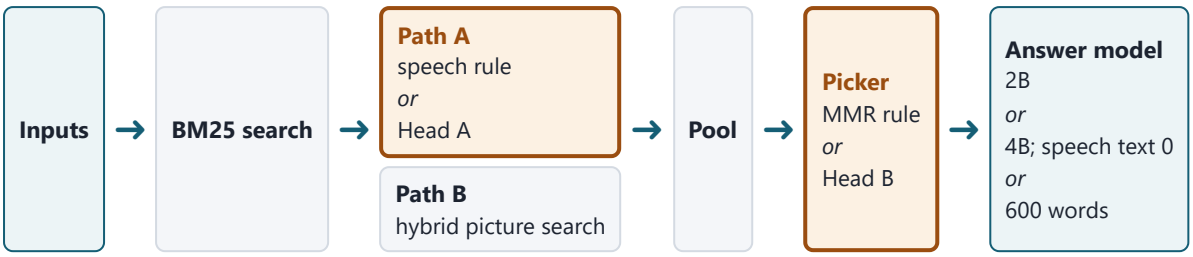
Important speech kept = the words include the speech within 10 s of the human-marked answer seconds. Open-ended answers graded by a free NVIDIA-hosted judge; it agreed with hand grading on 40 of 40 checked answers. Chosen policy vs v2 on open-ended: +8.5 (confirmed).

9 · Larger answer model (4B) on the test questions

System (4B unless stated)	Long videos (55)	Screen-reading (202)	Understanding (202)	All 459
Starting framework v2	38.2	60.9	43.1	50.3
Best framework, no speech text	41.8	62.9	45.5	52.7
Best framework + speech text	**49.1**	**67.8**	**45.0**	**55.6**
Dense MobileCLIP top-4 (standard rule)	40.0	54.8*	—	—
Best framework + speech text with the 2B model	49.1	49.0	31.7	41.4

Best + speech text vs starting framework (4B): +5.2 overall (confirmed); +6.9 on screen-reading lectures (confirmed); the speech text adds +2.8 over no speech text (confirmed). *188 of 202 questions (time budget ended).

10 · Full grid: every framework version × answer model × speech text



2B answer model, no speech text

Version	Long videos dev	Long videos TEST	Screen-reading dev	Screen-reading TEST	Understanding dev	Understanding TEST
Best framework	40.2	50.9	63.3	48.0	31.6	32.2
Best + Head A	39.0	41.8	62.2	45.0	32.7	33.7
Best + Head B	39.0	47.3	62.2	48.0	32.7	33.2
Best + Head A + Head B	40.2	34.5	62.2	52.0	30.6	32.7

2B answer model, with speech text (600 words)

Version	Long videos dev	Long videos TEST	Screen-reading dev	Screen-reading TEST	Understanding dev	Understanding TEST
Best framework	37.8	49.1	60.2	49.0	34.7	31.7
Best + Head A	37.8	40.0	58.2	48.0	29.6	33.2
Best + Head B	37.8	40.0	62.2	47.5	29.6	34.7
Best + Head A + Head B	36.6	34.5	60.2	51.0	30.6	32.7

4B answer model, no speech text

Version	Long videos dev	Long videos TEST	Screen-reading dev	Screen-reading TEST	Understanding dev	Understanding TEST
Best framework	36.6	41.8	65.3	62.9	39.8	45.5
Best + Head A	45.1	41.8	65.3	65.8	36.7	45.5
Best + Head B	40.2	38.2	64.3	64.4	43.9	48.5
Best + Head A + Head B	47.6	34.5	66.3	66.3	41.8	49.0

4B answer model, with speech text (600 words)

Version	Long videos dev	Long videos TEST	Screen-reading dev	Screen-reading TEST	Understanding dev	Understanding TEST
Best framework	40.2	49.1	68.4	67.8	38.8	45.0
Best + Head A	51.2	47.3	65.3	66.8	37.8	47.0
Best + Head B	46.3	43.6	67.3	64.9	38.8	45.5
Best + Head A + Head B	50.0	45.5	70.4	66.8	40.8	45.5

All combinations completed (research log steps 54–55). Test numbers in this grid beyond the declared comparison of section 7 are follow-up results.

With the **4B** model and no speech text, all three head versions score above the best framework on dev and test; Head A + Head B: dev +4.3, test +2.2, and **+3.0 over dev and test together (737 questions, confirmed)**. With the 600-word speech text the gain mostly disappears on test (+1.2 over dev and test, not confirmed). With the 2B model no head version differs reliably from the best framework.

11 · Summary: what gave what

Framework / component	What it achieved	On which data
v1 MobileCLIP top pictures	41.6% (vs 32.5% transcript only)	LongVideoBench
v3 best framework (rules, tuned)	50.9% vs 34.5% for the starting framework (+16.4, confirmed)	long everyday videos, test
Best framework + speech text + 4B model	**55.6% vs 50.3%** (+5.2, confirmed); 67.8% on screen-reading lectures (+6.9, confirmed)	all test questions; lectures
Speech-text selection	keeps 68% of the important speech (v2: 28%); open-ended +8.5 (confirmed)	long videos, dev
Head A (trained)	finds the answer moment 36.6% vs 28.0% (+8.5, confirmed); right picture among candidates 64.6% vs 59.8% (confirmed); accuracy not improved	long everyday videos
Head B (trained)	chosen pictures include the answer moment 50.9% vs 47.3% (with Head A: 56.4%); accuracy not improved	long everyday videos, test
Head A + Head B with the 4B model, no speech text	+3.0 over the best framework (dev and test together, 737 questions, confirmed); test alone 54.9% vs 52.7%	all benchmarks; lectures test 57.7% vs 54.2%
Confidence choice	best 2B overall row, 42.3% vs 41.4% (not confirmed)	all test questions

Best system for educational videos so far: **best framework + speech text + 4B answer model** (67.8% screen-reading, 45.0% understanding, test; 55.6% on all 459). The heads help the 4B model when it gets no speech text, but do not add to the best system once the speech text is given.